Database Design Specification: College Registration System

Design Activity & Practical: requirements, conceptual model, relational tables and data types (drawn in Draw.io)

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1. Purpose & scope

The system stores departments, programmes, courses and lecturers, and lets admitted students register for the courses in their programme each semester. The Registrar approves registrations and lecturers record final grades. *Out of scope:* fees, timetabling, admissions applications.

2. Users

Student	registers / drops courses, prints slip, views grades
Registrar	enrols students, maintains curricula, approves registrations
Lecturer	views class lists, enters final grades
Head of Dept.	assigns courses to lecturers, views statistics
Sys. Admin	user accounts, backups, security

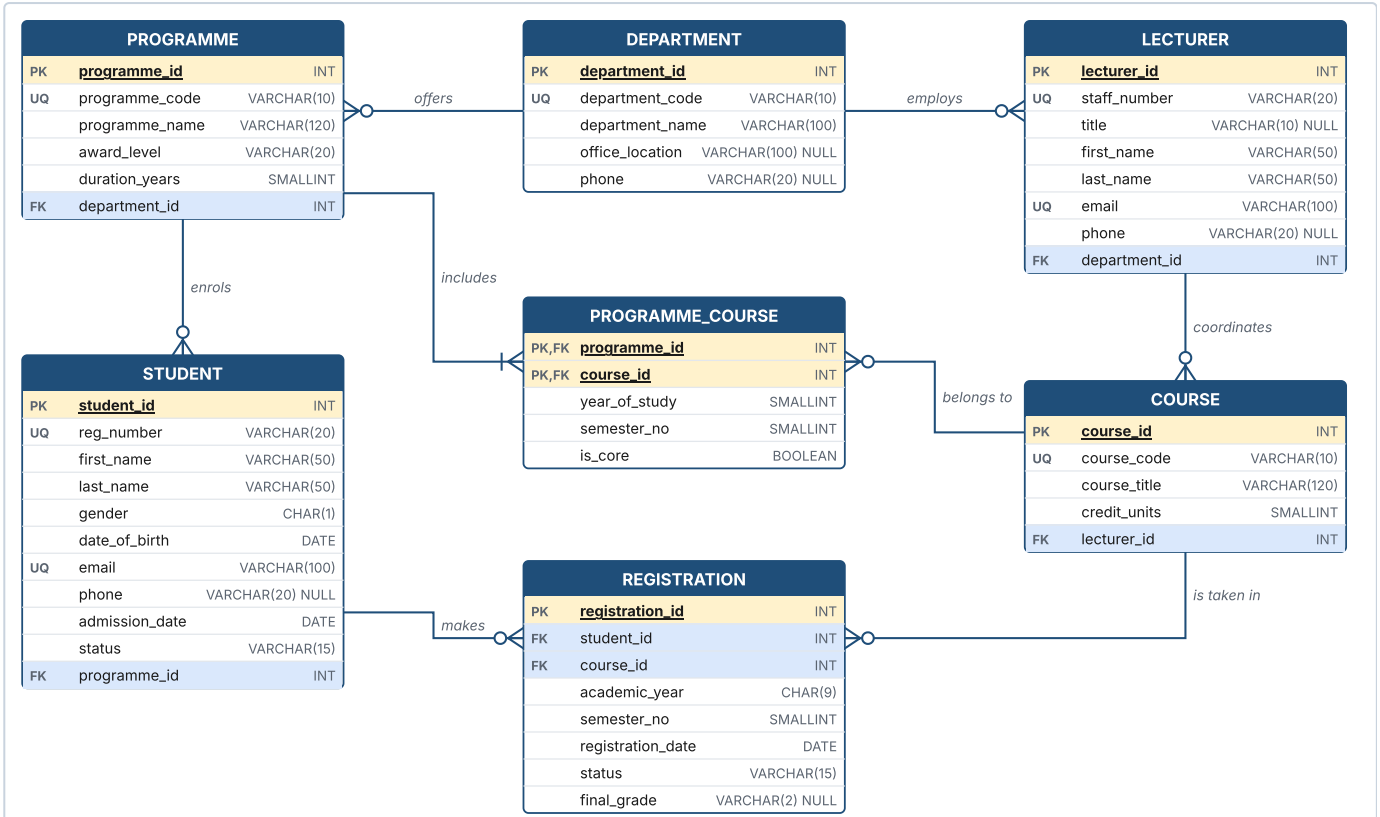
3. Processes

1. Set up departments, programmes and each programme's curriculum.
2. HoD allocates a coordinating lecturer to each course.
3. Registrar enrols a student into one programme (unique reg. number).
4. Student registers for courses each semester (status *Pending*).
5. Registrar approves the registration, or the student drops it during add/drop.
6. Lecturer records the final grade (status *Completed*).
7. Reports: class lists, registration slips, pending approvals.

4. Business rules

BR1	A department offers many programmes and employs many lecturers; each programme or lecturer belongs to **one** department.
BR2	A student is enrolled in exactly **one** programme.
BR3	A programme includes many courses, and a course can be in many programmes (M:N, resolved by `PROGRAMME_COURSE`).
BR4	Each course is coordinated by exactly one lecturer.
BR5	A student registers for many courses (M:N, resolved by `REGISTRATION`), but for the **same course only once per semester**.
BR6	Students may only take courses in their own programme, must be *Active*, and may take at most 24 credit units per semester.
BR7	Status is Pending, Approved, Dropped or Completed; a grade is only allowed when the status is Completed.
BR8	Reg. no., staff no., course and programme codes, and e-mails are unique.

5. Relational schema: 7 tables, keys, data types and relationships (crow's foot)



PK primary key | FK foreign key | UQ unique | |——< one-to-many | o< zero or many | |< one or many • Columns are NOT NULL unless marked NULL. The conceptual ERD (Chen notation) is on page 1 of the Draw.io file.

6. Design checks

- **Primary keys:** every table has one. Surrogate `INT` keys are used, except `PROGRAMME_COURSE`, which has the composite key (programme_id, course_id).
- **Relationships:** each 1:N is a NOT NULL FK on the "many" side, and each M:N is resolved by an associative table. FKs use `ON DELETE RESTRICT`.
- **Normalised to 3NF:** there are no repeating groups or partial dependencies. A student's department is derived through the programme, not stored again.

7. Data types & constraints

- `VARCHAR(n)` for names, codes and e-mails; `CHAR` for fixed lengths (`gender`, `academic_year` e.g. '2026/2027').
- `DATE` for birth, admission and registration dates; `SMALLINT` for credits, years and semesters; `BOOLEAN` for is_core.
- `CHECK`: semester in (1,2), credits 1–6, valid status and grade values. `UNIQUE` (student, course, year, semester) enforces BR5.
- Implemented and tested in `sql/01_schema.sql`.